

Khulna University of Engineering & Technology
Department of Urban and Regional Planning
BURP 3rd Year 2nd term Regular Examination, 2022
URP 3223
(Regional Development Planning)

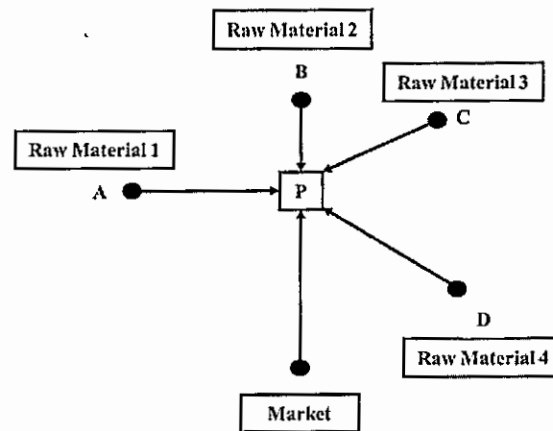
Full Marks: 210

Time: 3.00 hrs

- N.B.** i) Answer any three questions from each section in separate script.
ii) Figures in the right margin indicate full marks.

Section – A

1. (a) What is Agro-politan growth theory? Write down the components of Agro-politan growth theory. (10)
- (b) "Growth pole theory works as a policy tool for promoting regional balance" - how so? Discuss in brief. (10)
- (c) "Post-industrial period has become the period of tertiary industrial sector of economy"- do you agree? Explain. (15)
2. (a) "Polarization can lead towards backwash effect"- discuss with example. (10)
- (b) Suppose you are the owner of an industry P. You are going to relocate your industry. Which location are you going to choose and what are the causes for relocation according to the Weber's approach? (10)



- (c) "Polarization economies leads towards three types of agglomeration economies" – briefly explain with necessary examples. (15)
3. (a) "Mix-up economies can overcome the problems of polarization effect" - how so? Discuss in brief. (10)
- (b) Suppose the inter-industry flow of the product of two industries are given as follows: (25)

Production Sector	Consumption Sector		Domestic Demand	Total Output
	X	Y		
X	30	40	50	120
Y	20	10	30	60

Determine the coefficient matrix and test Hawkin's-Simon conditions for the viability of the system. If the domestic demand changes to 80 and 40 units respectively, what should be the gross output of each sector in order to meet the new demands?

4. (a) "The level of influence differs according to the three principles of central place theory" - Explain. (20)
- (b) What is production function? A firm can produce a product based on the following production function: (10)

$$Q = 10 K^{\frac{3}{4}} \cdot L^{\frac{1}{4}}$$

Calculate the average labor product when the firm employs 55 capital units and 19 labor units.

- (c) How does regional growth follow the expansion of the regional export base? (05)

Section – B

5. (a) What is a planning region and how does it differ from formal and functional regions in terms of its definable characteristics? (10)
- (b) How can flow analysis be used to identify the primary, secondary and tertiary functional areas of shopping? Identify it mathematically with an example. (15)
- (c) The distance between village A to town X and town Y is 4 km and 2 km respectively. Find the gravitational attraction of village A for both towns and compare the results drawing a figure. The number of retailers for Village A, Town X and Town Y are 50, 1000 and 200 respectively. [Here, the value of $K = 1.5$] (10)
6. (a) In what ways does access to energy and transportation and communication systems impact regional development and contribute to disparities? (10)
- (b) What strategies does a planner suggest to address regional disparities, particularly in lagging regions? (10)
- (c) Find out the regional disparity following Sopher Index on the basis of male and female literacy rate from given table. (15)

Districts	Dhaka	Chattogram	Sylhet	Khulna	Rajshahi	Mymensingh	Rangpur
Male literacy rate (%)	80.08	77.74	73.54	77.16	73.90	68.77	73.88
Female literacy rate (%)	76.02	75.73	70.39	72.95	69.97	65.49	67.69

Source: Population and Housing Census 2022

7. Dhaka, the primate city of Bangladesh, is mostly showing divergence tendencies since all the infrastructural and technological development is centered on it. But the periphery areas of Dhaka are showing convergence tendencies, and this trend is ensuring regional growth in the adjacent towns.
Read the aforementioned statement and answer the following questions.
 - (a) What factors contribute to Dhaka's dominant infrastructural and technological developments compared to other regions in Bangladesh? (10)
 - (b) How does this focused development in Dhaka influence the migration patterns within Bangladesh? (10)
 - (c) What policies would be effective in redistributing resources and opportunities to ensure balanced regional growth across Bangladesh? (15)
8. (a) Shortly discuss the different levels of regional planning with examples. (15)
- (b) Discuss the difference between place prosperity and people prosperity in regional policy. (10)
- (c) Give a short overview of different regional development policies in Bangladesh. (10)

Khulna University of Engineering & Technology
Department of Urban and Regional Planning
BURP 3rd Year 2nd Term Examination, 2022
URP 3291
(Operations Research and Systems Analysis)

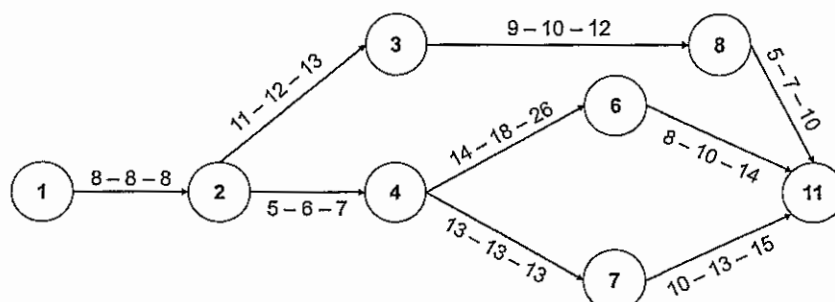
Full Marks: 210

Time: 3.00 hrs

- N.B.** i) Answer any **three** questions from each section in separate script.
ii) Figures in the right margin indicate full marks.
iii) Necessary charts and graphs will be provided.

Section – A

1. (a) What is Operations Research (OR)? Explain how studying OR can be helpful for effective decision making. (15)
- (b) A firm makes two products X and Y, and has a total production capacity of 9 tonnes per day. Both X and Y require the same production capacity. The firm has a permanent contract to supply at least 2 tonnes of X and at least 3 tonnes of Y per day to another company. Each tonne of X requires 20 machine hours of production time and each tonne of Y requires 50 machine hours of production time. The daily maximum possible number of machine hours is 360. All of the firm's output can be sold. The profit made is Rs 80 per tonne of X and Rs 120 per tonne of Y. Formulate this problem as an LP model- and solve it by using Graphical method- to determine the production schedule that yields the maximum profit. (20)
2. (a) What is linear Programming? In What types of real-world problems is linear programming commonly applied? (08)
- (b) Solve the following linear programming problem. Use big-M method. (17)
Minimize, $z = 3x_1 + 2x_2$
Subject to, $2x_1 + x_2 \geq 10$;
 $-3x_1 + 2x_2 \leq 6$;
 $x_1 + x_2 \geq 6$;
 $x_1, x_2 \geq 0$.
- (c) Define redundant constraints. How does the simplex method handle redundant constraints? (10)
3. (a) Consider the following LP model. Solve the problem using two-phase method. (20)
Minimize, $z = 9x_1 + 10x_2$
Subject to, $x_1 + 2x_2 \geq 25$;
 $3x_1 + 2x_2 \geq 60$;
 $x_1, x_2 \geq 0$.
- (b) Solve the following problem using simplex method. (15)
Maximize, $z = 6x_1 - 2x_2$
Subject to, $2x_1 - x_2 \leq 2$;
 $x_1 \leq 4$;
 $x_1, x_2 \geq 0$.
4. (a) What are meant by CPM and PERT? Describe how they help in controlling of a project. (10)
- (b) The following precedence diagram reflects three time estimates for each activity. (20)



- Determine; (i) The expected completion time for each path and its Variance.
(ii) The probability that the project can completed in 46 Weeks or less

(c) Define Gantt chart. How is it useful for project planning? (05)

Section – B

5. (a) What is game? Write down the characteristics of two persons zero sum game. (10)
(b) When can graphical method be applied for solving a game. (05)
(c) Solve the following game graphically. The Payoffs are for player A. (20)

		Player B	
		B ₁	B ₂
Player A	A ₁	2	- 3
	A ₂	- 4	2
	A ₃	6	4
	A ₄	- 3	1
	A ₅	5	0

6. (a) What is simulation? Write down some practical examples of a Simulation model. (13)
(b) A businessman has two independent investments A and B available to him but he lacks the capital to undertake both of them simultaneously and vice versa. The probability of success for A is 0.80 while for B is 0.50. Both investments require an initial capital outlay of \$4000 and both return nothing if the venture is unsuccessful. Successful completion of A will return \$3000 (over cost) and successful completion of B will return \$5000 (over cost). Draw the decision tree and determine the best strategy. (10)
(c) Consider the following pay-off (profit) matrix. (12)

	b ₁	b ₂	b ₃	b ₄	b ₅
a ₁	- 13	2	0	7	10
a ₂	5	10	- 15	5	8
a ₃	1	- 5	12	20	- 7
a ₄	3	14	2	- 12	0
a ₅	- 9	5	- 1	18	15

No Probabilities are known for the occurrence of future states. Determine the solution by the following criteria.

(i) Laplace; (ii) Maximin; (iii) Savage; (iv) Hurwicz (assume, $\alpha = 0.75$)

7. (a) Write down the applications of Markov chain. (05)
(b) An urn always contains 4 balls. The ball colors are red and blue. We can say that the State i , $i = 0, 1, 2, 3, 4$, if the urn contains i red balls. At each stage, a ball Randomly chosen and then replaced by new ball, which with a probability of 0.7 is the same color, and with a probability of 0.3 is the opposite color, as the ball it replaces. Find its transition probability matrix with detailed calculation. (18)
(c) A man's exercise habits are as follows. If he does his exercise one day, he is 80% sure not to do them the next day. On the other hand, if he does not do his exercise one day, he is \$ 30% sure not to do them the next day. In the long run, how often does he do his exercise? (12)
8. (a) The time Barber Joe takes to give a haircut is exponential with a mean of 10 minutes. Because of his popularity, customers usually arrive at a rate much higher than Joe can handle, 9 customers per hour which follows Poisson distribution. Joe really will feel comfortable if the arrival rate is effectively reduced to about 5 customers per hour. To accomplish this goal, he came up with the idea of providing limited seating (15)

in the waiting area, so that newly arriving customers will go elsewhere when they discover that all the seats are taken. How many seats should Joe provide to accomplish his goal?

- (b) Mc Burger fast food restaurant has only one Cashier. Customers arrive according to a Poisson distribution every 7 minutes and form one line to be served. The time to fill an order is exponentially distributed with a mean of 10 minutes. The waiting line does not accommodate more than 10 customers, including the one being served. Arriving customers go elsewhere if the line is full. Determine the following: (20)
- (i) Probability that the cashier is busy;
 - (ii) Probability that an arriving customer will not get service immediately but will find an empty space in the line;
 - (iii) Percentage of customers that will not get food from the restaurant;
 - (iv) Suppose that to three customers would like to get food together. What is the probability that their wish can be fulfilled?

Khulna University of Engineering & Technology
Department of Urban and Regional Planning
BURP 3rd Year 2nd Term Regular Examination, 2022
URP 3231
(Transportation Planning)

Full Marks: 210

Time: 3.0 hrs

- N.B.** i) Answer any three questions from each section in separate script.
ii) Figures in the right margin indicate full marks.

Section – A

1. (a) The following table gives the particulars collected for a section of road in Khulna 1.6 km long during the course of a moving observer study: (22)

Direction of the Run	Journey Time (min)	Stopped Time (min)	Vehicles met with in the opposing direction			Vehicles in the same direction	
			CNG	Bus	Mahindra	Over-taking vehicles	Overtaken vehicles
N-S	1.01	0.04	13	0	5	1	0
S-N	0.92	0.10	7	1	6	0	1
N-S	0.77	0.08	12	2	14	4	3
N-S	1.03	0.14	7	2	5	5	2
S-N	0.84	0.08	9	0	13	2	0
S-N	1.05	0.19	5	1	18	1	0
N-S	1.05	0.16	4	2	8	3	1

Calculate the flow in PCU/hr in both directions (N-North & S- South) of traffic assuming an equivalency factor of 3 for bus and 1.5 for CNG & mahindra. Calculate the Journey Speed and Running Speed.

- (b) Suppose, you are assigned to introduce local bus transport in Khulna. Discuss the role of different agencies in bus transportation that will involve in the process. (13)
2. The Dhaka metro rail system was expected to connect key areas of Dhaka, providing a faster and more efficient mode of transportation for the residents. Passengers are getting on and off the metro rail at all 16 stations on the Uttara-Motizheel route.

- (a) Make a comparative overview among types of walking that can occur in these metro stations. (09)
- (b) Find out the time-space demand and supply for pedestrian circulation of a Mezzanine of Dhaka University Metro station from the following information: (17)

Types of Movement (per hour)	No. of Persons	Time (Sec)
Stair to turnstiles	1700	17
Stairs to token booth to turnstiles	700	55
Stairs to newsstand to turnstiles	170	58
Turnstiles to newsstand to stairs	100	45

Assume that the area of the mezzanine is 1800 sq.ft including an unused area of 300 sq.ft and expected LoS is C/D.

- (c) Briefly elaborate the factors with example that are considered for travel demand modeling for Dhaka metro rail project. (09)

3. (a) What is meant by time-mean speed and space-mean speed? Explain with necessary examples. (10)
- (b) It is observed that on an average a vehicle driver drives 1700 km during the course of a year. The probability of having an accident is 17 per 17 million vehicle-kilometers. What is the probability of a driver having at least two accidents during his driving career extending to 17 years? (10)
- (c) What are the factors that influence the required bus fleet strength of a city? Calculate the required fleet strength of buses for Khulna city from the following data set: (15)

Population: 3, 050,375;
 Per Capita Trip Rate: 1.8;
 Share of bus trips in overall trips: 15%;
 Average trip length: 6.2 Km;
 Distance travelled by bus per day: 230 km;
 Capacity of Bus: 60 passengers;
 Average occupancy: 0.85;
 Expected fleet utilization: 75%.
4. (a) Define Traffic Accident? Mention types of Collision with necessary sketches. Also, Mention methods available for analyzing accident data. (10)
- (b) What categories are there for city bus routes? Mention the bus route planning principles. (11)
- (c) "TSM are more efficient than TDM" - Provide pertinent strategies and examples to support your argument. (14)

Section – B

5. (a) Explain the concept of a transportation system. Discuss in brief the components of the transportation system that you frequently use. Do you think the system is perfect? If so, describe the strength of the system. If not, explain why. (10)
- (b) Suppose you are responsible for planning a new expressway from Dhaka to Mongla port. What basic elements of transportation planning will you consider? Discuss in brief with necessary examples. (15)
- (c) What is Traffic Analysis Zone (TAZ)? Mention the selection criteria of TAZ. (10)
6. (a) Discuss the following with necessary example: (20)
 - i) Influential factors of travel demand
 - ii) Travel Analysis Zone
 - iii) Origins and Destinations
 - iv) Productions and Attractions
- (b) Describe regression and cross-classification methods of trips production with necessary examples. (15)

7. (a) 5 Thana in Khulna City have the following information:

(15) 20

TAZ	Thana name	Production	Attraction
1	Kotwali	45822	35351
2	Khan Jahan Ali	8740	7601
3	Khalishpur	33550	42681
4	Daulatpur	15668	37253
5	Sonadanga	34983	15877

The travel matrix is:

TAZ	Travel Cost (BDT)				
	1	2	3	4	5
1	35	10	25	15	50
2	30	50	25	40	15
3	5	35	10	30	30
4	10	25	5	25	25
5	30	15	25	10	40

Travel Cost (BDT)	F_{ij}	Travel Cost (BDT)	F_{ij}
5	5.55	30	1.75
10	3.50	35	1.01
15	4.87	40	0.66
25	2.59	50	0.32

Apply the gravity model to distribute the trips ($K_{ij}=2.25$).

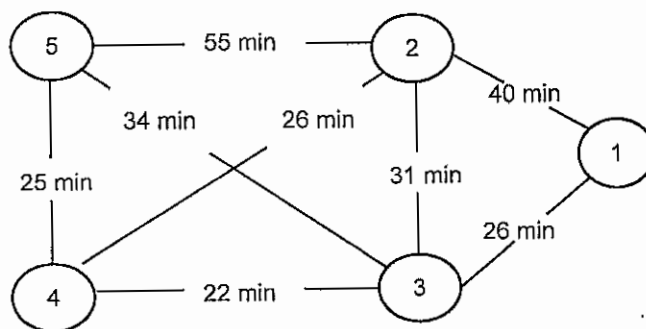
- (b) Discuss the following of trip distribution analysis: (15)
- Uniform Growth Factor
 - Singly-Constrained Growth Factor
 - Average Factor
8. (a) Let the number of trips from 'i' to 'j' is 5000, and three modes are available (15)
which has the following characteristics.

	Vehicle travel time between 'i' and 'j'	Walking time to and from stops	Waiting time at stops	Fare charged to travel between 'i' and 'j'	Parking cost
Co-efficient	0.09	0.07	0.14	0.39	0.14
Car	20	-	-	4	18
Bus	30	3	5	-	6
Train	12	2	10	-	4

Decide the best alternative in terms of trips carried and compute the fair required to travel by each mode.

- (b) A trip table (veh/hr) needs to be loaded on the network shown in figure. Find (20)
the new link travel time using capacity restrained assignment. Assume capacity of each link to be 5750 veh/hr.

To \ From	1	2	3	4	5
1	0	475	570	665	285
2	380	0	285	570	760
3	855	380	0	190	475
4	760	665	855	0	285
5	285	380	475	570	0



Khulna University of Engineering & Technology
Department of Urban and Regional Planning
BURP 3rd Year 2nd Term Regular Examination, 2022
CE 3261
(Water Resources Planning)

Full Marks: 210

Time: 3.00 hrs

- N.B.** i) Answer any three questions from each section in separate script.
ii) Figures in the right margin indicate full marks.

Section – A

- 1 (a) Define hydrologic cycle. Describe the main components of hydrologic cycle with neat sketch. (08)
- (b) Describe residence time with suitable example and prove that the residence time for global river is 17.3 day. (07)
- (c) The following observations over period of one month were made and shown in table 1(c) for conducting water balance of a reservoir that was constructed by KUET administration in KUET campus. (10)
-Write down that water-budget equation for the reservoir.
-Calculate the pan evaporation coefficient.

Average surface area= 10 km ²	Mean surface outflow rate= 15 m ³ /s
Mean surface inflow rate= 10 m ³ /s	Pan evaporation= 200 mm
Drop in reservoir level= 1.5 m	Average seepage discharge= 164 mm
Rainfall= 100 mm	

- (d) Write down the sources of hydrological and metrological data in Bangladesh. Show that rainfall run-off process on a watershed is a hydrologic system. (10)
- 2 (a) What are the analytical methods of evaporation estimation? Describe the energy-budget method of evaporation estimation. (12)
- (b) Calculate the potential evapotranspiration from 'KUET Padma Pukur' in March month for the following information- (13)

Latitude	22°5' North
Mean monthly temperature	28°C
Mean relative humidity	63%
Mean observed sunshine hours	10.5 h
Wind velocity at 2 m height above ground	125 km/d
Nature of surface cover	Water surface

- (c) Define the following terms: (10)
i. Aquiclude
ii. Flowing Artesian well
iii. Perched water table
iv. Draw down
v. Coefficient of permeability
- 3 (a) Define unit hydrograph. Write down the factors affecting flood hydrograph. (05)
- (b) A 6-hour storm occurs over 100 km² watershed. The details of the catchment are shown in table. Calculate the run-off volume from the catchment and the hourly distribution of the effective rainfall for 100 km² catchment area. (15)

Sub Area (km ²)	Φ-index (mm/hr)	Hourly Rain (mm)					
		1 st hour	2 nd hour	3 rd hour	4 th hour	5 th hour	6 th hour
35	20	10	42	14	10	8	4
25	15	12	42	14	18	15	9
10	15	25	33	18	21	8	5
15	10	16	48	22	10	18	10
05	16	15	42	25	12	18	8
10	15	22	31	18	20	10	5

- (c) In a typical 4-hour storm producing 5 cm of excess rainfall from a basin, the following flows in the stream are recorded and shown in table. (15)

Estimate the peak flow and time of its occurrence in a flood created by 12-hr storm, which produces 2.5 cm of effective rainfall during first 4-hrs, 3.3 cm effective rainfall during second 4-hrs, and 3.7 cm effective rainfall during third 4-hrs.

Time (hr)	0	2	4	6	8	12	16	20	24	28	32	36	40
Flow (m ³ /s)	0	25	90	130	190	150	110	75	45	20	8	3	0

- 4 (a) Write down the considerations for the placement of a rain gauge. Draw and describe a non-recording rain gauge. (10)
- (b) For the confined flow into a well at the edge of the zone of influence, show that, (10)
- $$Q = \frac{2\pi T S w}{\ln \frac{R}{rw}}$$
- (c) Analysis of data on maximum rainfall depth at Khulna indicated, a depth of 290 mm had a return period of 60 years. Determine the probability of one day rain fall depth equal to or greater than 290 mm at Khulna occurring (15)
- Once in 50 years successively
 - Three times in 40 years successively

Section – B

- 5 (a) Define “Duty” and “Delta”. Derive the relationship between duty and delta. (10)
- (b) What is conjunctive use of water in irrigation and what are the advantages and disadvantages? (06)
- (c) What is permanent wilting point? In one irrigated field, the observed depths at 10 stations are 2.5, 2.6, 2.7, 2.9, 3.2, 1.6, 1.7, 1.8, 1.9, 2.1 meters, respectively. Find out the water distribution efficiency. (06)
- (d) While designing an irrigation pump system, it was found out that, (13)
- Field capacity of the soil = 30%
 - Permanent wilting point = 16%
 - Dry density of soil = 1.4 gm/cc
 - Effective depth of root zone = 70 cm
 - Daily consumptive use of water for the given crop = 15 mm
- So, after how many days will you supply water to the soil?
- 6 (a) Classify the river training works for discharge and explain briefly. (06)
- (b) Draw a neat sketch for a typical levee. (06)
- (c) The following data was collected for designing a guide bank. (23)
- Maximum discharge = 8500 m³/s

Highest flood level = 105.25 m
 River bed level = 100.25 m
 Average diameter of river bed material = 0.15 mm
 Design and neatly sketch any of the cross section for the guide bank.

- 7 (a) Define irrigation and irrigation engineering. Discuss the application of irrigation engineering with regards to climate change perspective in Bangladesh. (10)
- (b) Determine the classification of irrigation water having the following characteristics: (09)
 Concentration of Na, Ca, and Mg are 32, 8, and 4 milli-equivalent per liter respectively; and the electric conductivity is 900 $\mu\text{S}/\text{cm}$ @ 25°C.
- (c) Differentiate between (16)
 i. Fluvial system and drainage basin
 ii. Rocky River stage and Boulder stage
 iii. Perennial River and non-Himalayan River
 iv. Meandering channel and Braided channel
- 8 (a) How and why secondary currents are formed in a bend and how this affects shoal formation at a convex edge of a bend? (10)
- (b) What are spur dykes? Explain with neat sketches, the types of groynes according to alignment. (12)
- (c) For a river the estimated flood of a two return periods are given below. Find out the discharge of floods that have a return period of 300 years and 1000 years. (13)

Return Period (Years)	Peak Flow (m^3/s)
250	650
90	410

Table 01: Slope of the Saturation Vapour Pressure vs Temperature Curve at the Mean Air Temperature, in mm of Mercury per °C

Temperature (°C)	A (mm/°C)
0	0.30
5.0	0.45
7.5	0.54
10.0	0.60
12.5	0.71
15.0	0.80
17.5	0.95
20.0	1.05
22.5	1.24
25.0	1.40
27.5	1.61
30.0	1.85
32.5	2.07
35.0	2.35
37.5	2.62
40.0	2.95
45.0	3.66

Table 02: Mean Monthly Solar Radiation at Top of the Atmosphere in mm of Evaporable Water / Day

North latitude	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
0°	14.5	15.0	15.2	14.7	13.9	13.4	13.5	14.2	14.9	15.0	14.6	14.3
10°	12.8	13.9	14.8	15.2	15.0	14.8	14.8	15.0	14.9	14.1	13.1	12.4
20°	10.8	12.3	13.9	15.2	15.7	15.8	15.7	15.3	14.4	12.9	11.2	10.3
30°	8.5	10.5	12.7	14.8	16.0	16.5	16.2	15.3	13.5	11.3	9.1	7.9
40°	6.0	8.3	11.0	13.9	15.9	16.7	16.3	14.8	12.2	9.3	6.7	5.4
50°	3.6	5.9	9.1	12.7	15.4	16.7	16.1	13.9	10.5	7.1	4.3	3.0

Table 03: Mean Monthly Values of Possible Sunshine Hours

North latitude	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
0°	12.1	12.1	12.1	12.1	12.1	12.1	12.1	12.1	12.1	12.1	12.1	12.1
10°	11.6	11.8	12.1	12.4	12.6	12.7	12.6	12.4	12.9	11.9	11.7	11.5
20°	11.1	11.5	12.0	12.6	13.1	13.3	13.2	12.8	12.3	11.7	11.2	10.9
30°	10.4	11.1	12.0	12.9	13.7	14.1	13.9	13.2	12.4	11.5	10.6	10.2
40°	9.6	10.7	11.9	13.2	14.4	15.0	14.7	13.8	12.5	11.2	10.0	9.4
50°	8.6	10.1	11.8	13.8	15.4	16.4	16.0	14.5	12.7	10.8	9.1	8.1

Khulna University of Engineering & Technology
Department of Urban and Regional Planning
BURP 3rd Year 2nd Term Regular Examination, 2022
CE 3207
(Elements of Civil Engineering Structures)

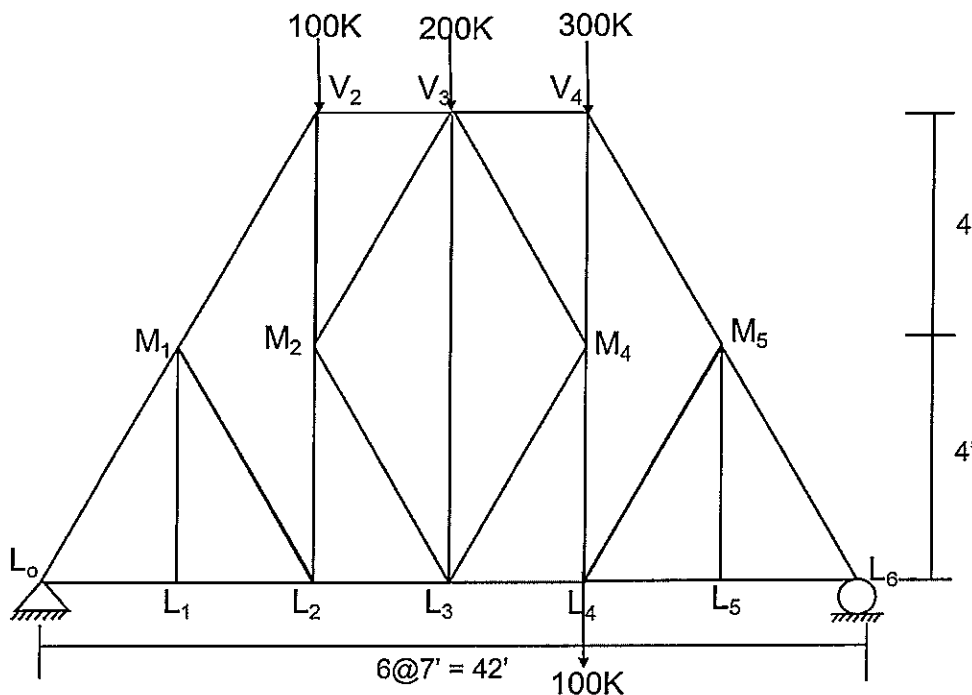
Full Marks: 210

Time: 3.00 hrs

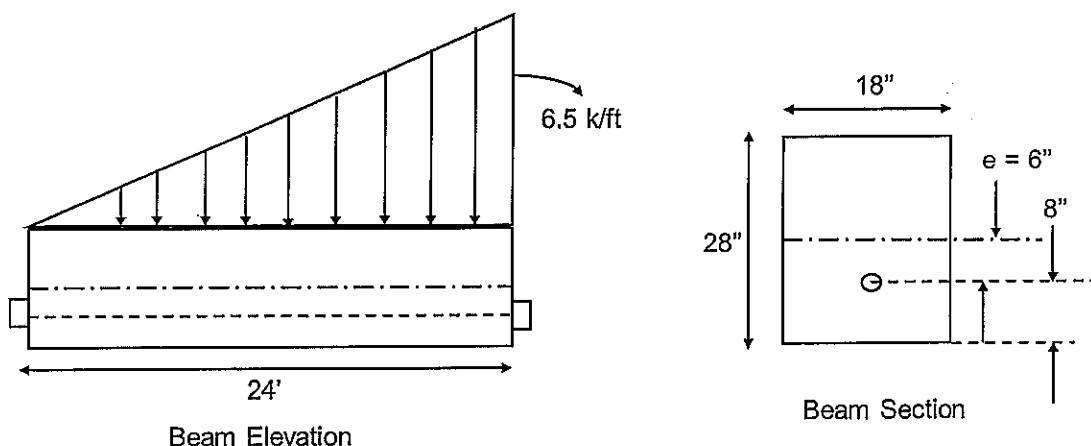
- N.B.** i) Answer any three questions from each section in separate script.
ii) Figures in the right margin indicate full marks.
iii) Necessary tables are Provided

Section – A

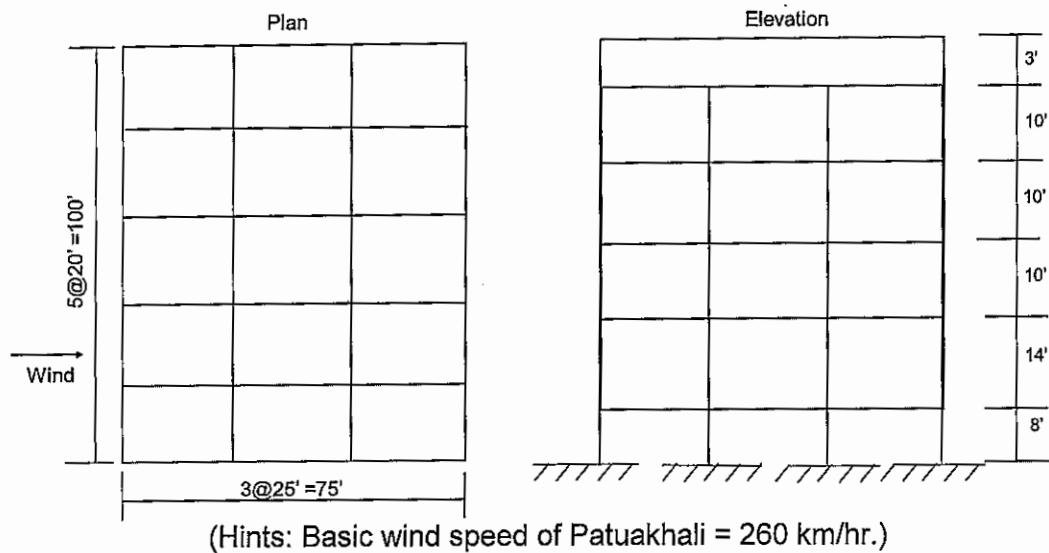
- 1 (a) Calculate the forces in member M_1V_2 , M_1L_2 , V_2V_3 , V_2M_2 and V_3L_3 of the following (35)
truss shown in figure.



- 2 (a) Define i) Prestressed concrete ii) Pre-tensioning iii) Post-tensioning (09)
(b) Describe the principle of prestressed concrete and compare it with the ordinary (10)
reinforced concrete.
(c) For the following pre-stressed rectangular beam, calculate the maximum stresses (16)
of top and bottom fiber in the concrete due to pre-stress force of 400k.



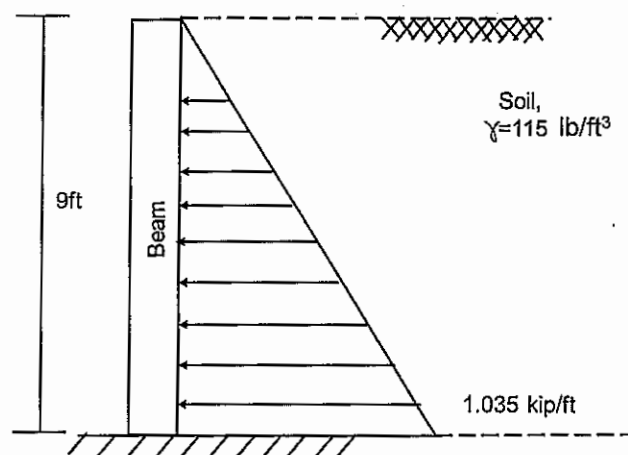
- 3 (a) The plan and elevation of a 4-storied reinforced concrete building is shown in figure (27)
below. It is to be built in the Patuakhali in an area that is classified as exposure
category C. Calculate-
1. Wind pressure distribution on building
 2. Total base shear and total overturning moment
 3. Factor of Safety against overturning if total dead load is 200 psf.



- (b) Write a short note on: (08)
- 1) General shear failure
 - 2) Local shear failure
- 4 (a) Define column. How can you fix the tie bar spacing in a tied column? (06)
- (b) Design a circular, spiral reinforced column for an allowable working load of 450 kips. Where, $f'_c = 3.5$ ksi and $f_y = 60$ ksi. (29)

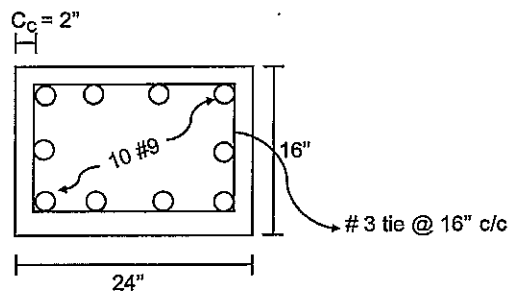
Section – B

- 5 (a) What are the constituents of reinforced concrete? Briefly describe the function of them. (10)
- (b) Draw the stress-strain diagram of mild steel and show all the limiting stress point and discuss them briefly. (10)
- (c) Cantilever beam with a length of 9ft supports a uniformly varying load of 1.035 k/ft and its own self weight as shown in the figure below. $f'_c = 3000$ psi and $f_y = 72500$ psi. Design the beam and draw the x-section and long section of the beam. (15)

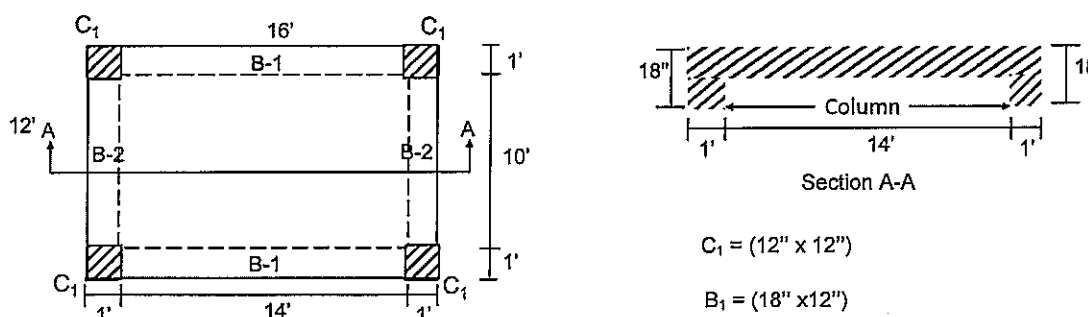


- 6 (a) How is safety provided in WSD method? Why does BNBC follow WSD method? (10)
- (b) Why is corer reinforcement provided? Describe the design procedure of it with neat sketch. (08)
- (c) A one-way simply supported slab is built to supports a uniform livelihood of 150 psf in addition to its self-weight. The clear span of the slab is 15ft. Use $f'_c = 3000$ psi and $f_y = 60000$ psi. Design the slab, following WSD method. (17)

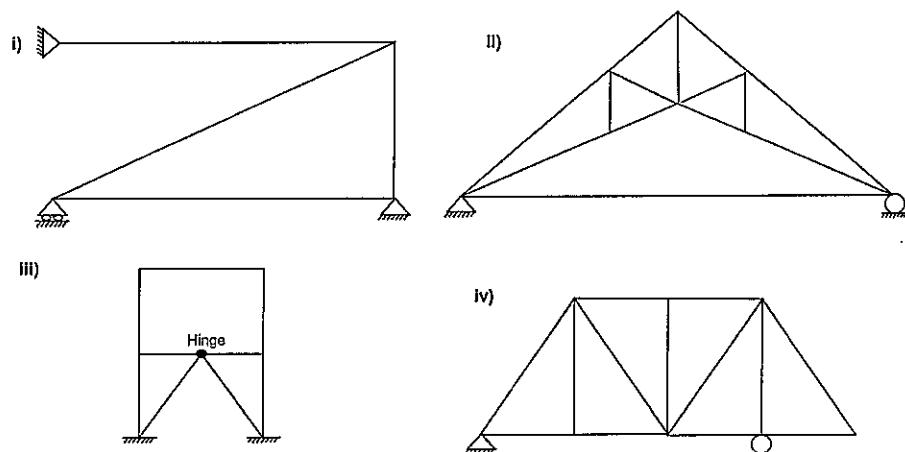
- 7 (a) Why is tie bar provided in column? (05)
- (b) A column with a cross-section shown in figure below is loaded concentrically. Calculate the allowable axial load of the column. Also check the requirement of extra ties in the column. $f'_c = 3000$ psi and $f_y = 60000$ psi. (12)



- (c) A 16'x12' slab has a thickness of 5.5in. It is supported by four beams having a width and depth of 12" and 18", respectively. The column supports the beam having a dimension of 12"x12". The slab carries a live load of 100 psf in addition to its self-weight. How much design load each beam and column will carry? (18)



- 8 (a) What are the different types of losses in prestressed concrete? Explain each of them briefly. (10)
- (b) Find out the determinacy and stability of the structures shown below. (10)



- (c) Calculate the earthquake load at each floor level of a typical 6 storied hospital building having equal height of 3.5m in Sylhet city. (15)

Floor area= 30m x 45m

Structural importance co-efficient = 1.0

Response modification co-efficient =1.2

The building is located at seismic zone ($z = 0.075$)

Steel building frame ($C_t = 0.083$)

Seismic load (dead load) = 18000 KN/floor